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DPAPI and DPAPI-NG: Decrypting All Users' Secrets and PFX Passwords'

Paula Januszkiewicz

CEO and Founder

CQURE

Level: Intermediate

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DPAPI and DPAPI-NG:

Decrypting All Users' Secrets and PFX Passwords

TechMentor, 12th August 2025

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DPAPI and DPAPI-NG:
Decrypting All Users' Secrets and PFX Passwords

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Microsoft Enterprise Security MVP

Microsoft Regional Director

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May 31st - June 2nd
Oslo Spektrum

June 1st | 10:00 CEST
Adventures in the Underland: Uncommon Hacker's Persistency Methods and Countermeasures

June 1st | 16:00 CEST
Hackers' Perspective on Remote Working: Know the Risks, Fight the Tricks

HACK IN PARIS CYBERSECURITY EVENT | 2022

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We are proud to announce that
Paula Januszkiewicz
was rated as
No 1 Speaker
at Microsoft **Ignite!!!**

May 4-8, 2015
Chicago, IL

No.1 Speaker

Paula Januszkiewicz
CEO CQURE

She received
a **"Best of Briefings"** award at her
"CQTools: The New Ultimate Hacking Toolkit"
Black Hat Asia 2019 briefing session

black hat

LEAP

RIYADH FRONT EXPO CENTRE, SAUDI ARABIA

Keynote: Cyber Attacks & Defense Transformation: The Look to Past, Present & Future.
by Paula Januszkiewicz, CEO of CQURE
February 3rd, 2022 | 16:25 AST (GMT +3)

Dr. Makaziwe...
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Ballon d'Or winning...

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Strong Story to Tell: Top 10 Mistakes by Administrators About Remote Work

Paula Januszkiewicz
CEO, Cybersecurity Expert, CQURE Inc.

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1. Consulting Services:
   a) Extensive IT Security Audits and Penetration Tests of all kinds,
   b) Configuration Audit and Architecture,
   c) Design Social Engineering Tests,
   d) Advanced Troubleshooting and Debugging,
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2. R&D & CQLabs Tools & Hacks Publications.

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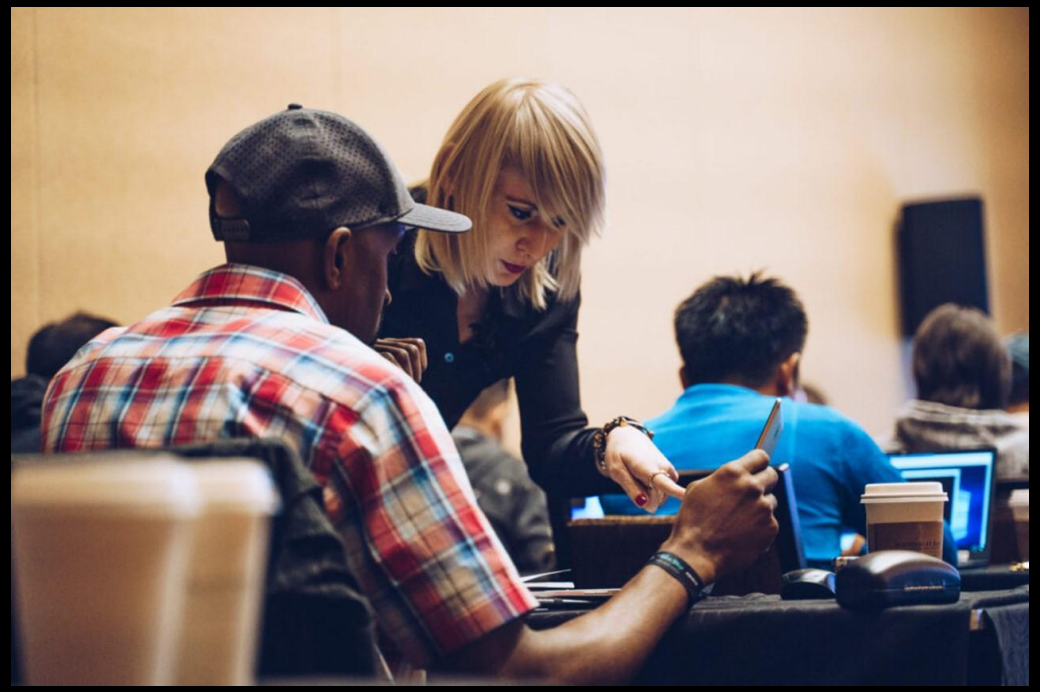

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Insightful Stats for 2025

$4.88M

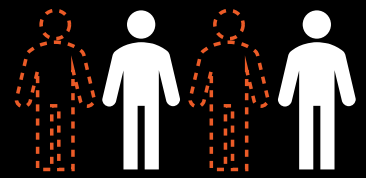
is the global average cost of a data breach in 2024.
It increased 10% over the previous year.

292 Days

that's how long, on average, it took to identify and contain breaches involving stolen or compromised credentials.

68%

of data breaches involved human error in 2024.



>50%

of breached organizations are facing high levels of security staffing shortages. That is a 26.2% increase from the prior year.

Source: IBM (2024), Verizon (2024)

Impactful Hacking Stats for 2025

43%

of all cyberattacks are aimed at small businesses.

95%

of data breaches are financially motivated.

83%

of breaches involve external actors.

64%

of organizations do not have a cyber security incident response plan.

Source: Verizon (2023); Accenture (2023), Kroll (2022)

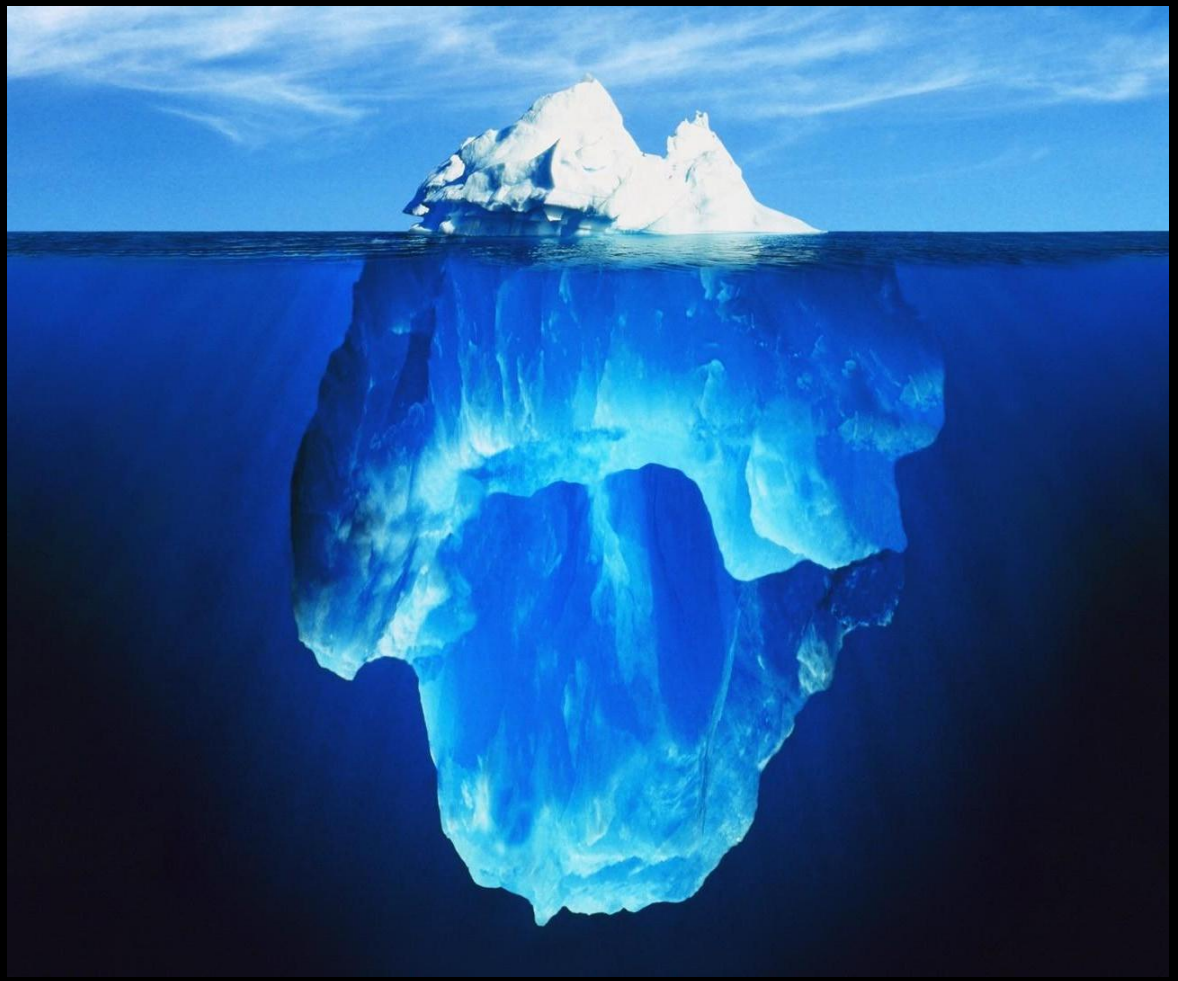
Data Protection API

CQURE

Windows Most Often Uses Following Algorithms

AES256-CBC (data encryption)
AES256-CCM (BitLocker)
AES256-GCM (DPAPI blobs)
AES128-ECB (BitLocker)
AES256-XTS (BitLocker)
RC4
MD4, MD5, SHA*
DES, 3DES
RSA 1024, RSA 2048 (used more often)
Secret agreement: Diffie-Hellman

DPAPI-NG:
Key derivation: SP800_108_CTR_HMAC (SHA512), KDF_SP80056A_CONCAT (client)



System Secrets

Bootkey

Class names for keys from HKLM\SYSTEM\CCS\Control\Lsa

Data
GBG
JD
Skew1

SAM/NTDS.dit
(MD4 Hashes)
C:\windows\system32\config
C:\windows\system32\NTDS

LSA Secrets
(Service Accounts)
HKLM\SECURITY\Policy\Secrets

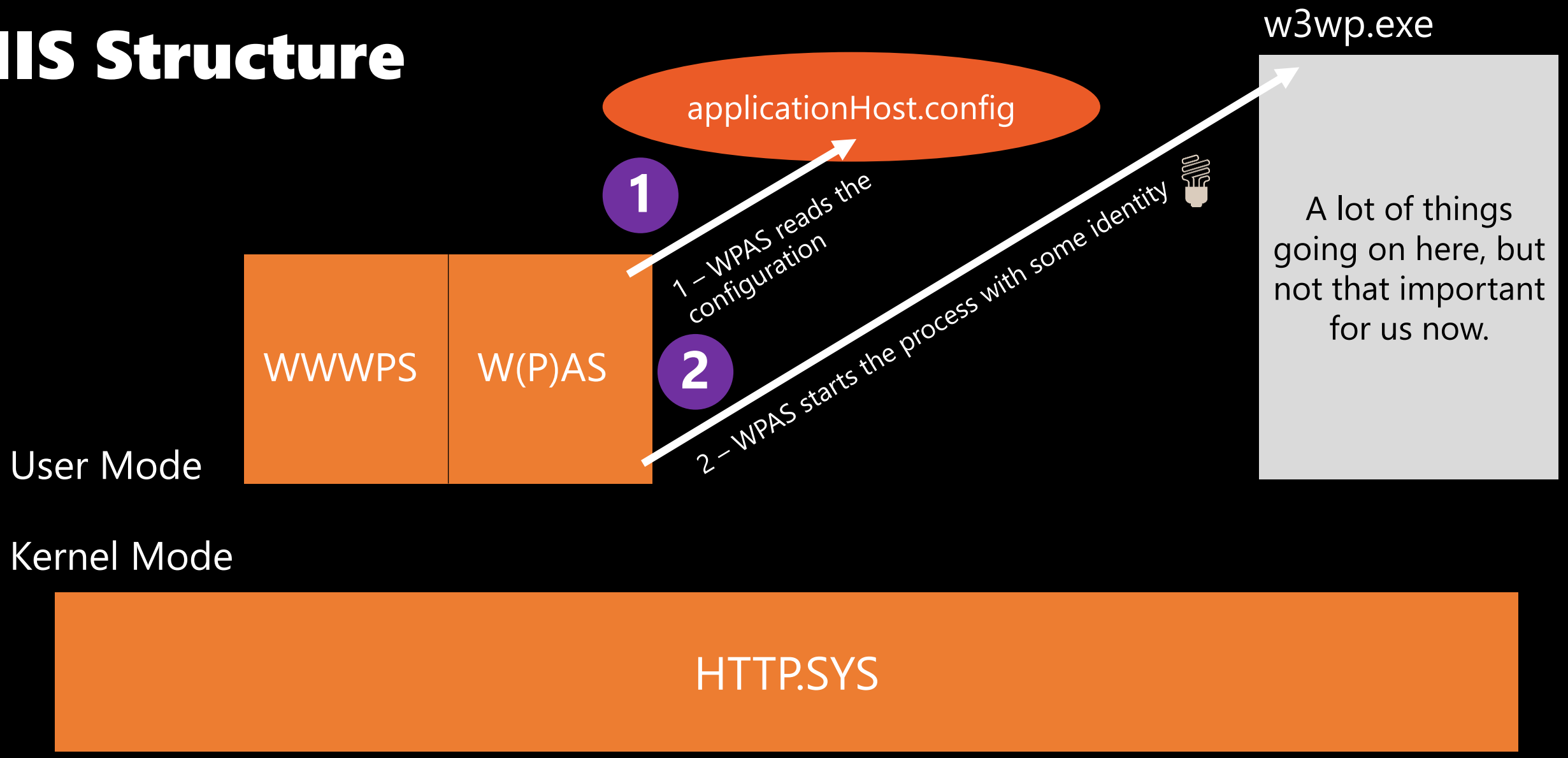
MSDCC2
(Cached Logon Data)
HKLM\SECURITY\Cache

$MACHINE.ACC
(SYSTEM's Clear Text Password)

DPAPI_SYSTEM (Master Keys)
HKLM\SECURITY\Policy\Secrets

More information: http://cqureacademy.com/blog

IIS Structure



IIS Configuration

- In contrast to the earlier IIS versions, IIS 10.0 is set to use two new Cryptography API: Next Generation (CNG) providers by default:

IISWASOnlyCngProvider and IISCngProvider. We still have: IISWASOnlyRsaProvider, AesProvider, IISWasOnlyAesProvider and RsaProtectedConfigurationProvider, DataProtectionConfigurationProvider

CNG stores shared private keys in the *%ALLUSERSPROFILE%\Application Data\Microsoft\Crypto\Keys*

- Worker Processes (w3wp.exe)

Their identity is defined in Application Pool settings

Are managed by Windows Process Activation Service that knows how to read secrets

- Passwords for AppPool identity can be 'decrypted' even offline

They are stored in the encrypted form in applicationHost.config

Conclusion: IIS relies it's security on Machine Keys (Local System)

IIS Configuration / Application Pools

- Used to group one or more Web Applications

Purpose: Assign resources, serve as a security sandbox

- Use Worker Processes (w3wp.exe)

Their identity is defined in Application Pool settings
Process requests to the applications

- Passwords for AppPool identity can be 'decrypted' even offline

They are stored in the encrypted form in applicationHost.config

Conclusion: IIS relies it's security on Machine Keys (Local System)

NTDS.DIT, SAM

To perform an analysis on NTDS.DIT the following information sources are needed from the domain controller:

- NTDS.DIT
- Registry hives (at least the SYSTEM hive)

SAM, ntds.dit are stored locally on the server's drive

They **do not** contain Passwords

They use **MD4** as a way of storing them

They are encrypted

The above means:
To read the clear text password you need to struggle!

Getting the: Hash

SAM

1. bootkey: classes from HKLM\SYSTEM\CCS\Control\Lsa + [class names for: Data, GBG, JD, Skew1] (+arrays' permutations)
2. F: HKLM\SAM\SAM\Domains\Account\ [F - value] string aqwerty = "!@#$%^&*()qwertyUIOPAzxcvbnmQQQQQQQQQQQQ)(*@&%\0"; string anum = "0123456789012345678901234567890123456789\0";
3. rchbootkey: MD5(string created after arytmetic functions with F, aqwerty, anum, bootkey)
4. hbootkey: RC4(key, data) -> RC4(rchbootkey, F)
5. MD5(…,hbootkey,…) -> RC4(…)-> DES(…, F) to get the hash (MD4)

Services

Store configuration in the registry

Always need some identity to run the executable!

Local Security Authority (LSA) Secrets

Must be stored locally, especially when domain credentials are used
Can be accessed when we impersonate to Local System

Their accounts should be monitored

If you cannot use gMSA, MSA, use subscription for svc_ accounts (naming convention)

Conclusion: Think twice before using an Administrative account, use gMSA

Getting the: Service Account Secrets

```
1. bootkey: classes from HKLM\SYSTEM\CCS\Control\Lsa + [class
names for: Data, GBG, JD, Skew1] (+permutations)
int[] permutationBootKey = new int[] { 0x8, 0x5, 0x4, 0x2,
0xb, 0x9, 0xd, 0x3, 0x0, 0x6, 0x1, 0xc, 0xe, 0xa, 0xf, 0x7 };
2. PolEKList: HKLM\SECURITY\Policy\PolEKList [default value]
3. lsakey: AES_DECRYPT(key, data) -> AES(bootkey, PolEKList)
4. NL$KM secret: HKLM\SECURITY\Policy\Secrets\NL$KM
5. nlkm_decrypted: AES_DECRYPT(lsakey, NL$KM secret)
```

User Secrets

Classic Data Protection API

- Based on the following components:

Password, data blob, entropy

- Is not prone to password resets!

Protects from outsiders when being in offline access
Effectively protects users data

- Stores the password history

You need to be able to get access to some of your passwords from the past

Conclusion: OS greatly helps us to protect secrets

Getting the: Classic DPAPI Secrets

DPAPI (classic)

A. MasterKey

1. pwdhash = MD4(password) or SHA1(password)
2. pwdhash_key = HMACSHA1(pwdhash, user_sid)
3. PBKDF2(…, pwdhash_key,…), another elements from the file. Windows 10 no domain: SHA512, AES-256, 8000 rounds
4. Control - HMACSHA512

B. CREDHIST

1. pwdhash = MD4(password) or SHA1(password)
2. pwdhash_key = HMACSHA1(pwdhash, user_sid)
3. PBKDF2(…, pwdhash_key,…), another elements from the file. Windows 10 no domain: SHA512, AES-256, 8000 rounds
4. Control - HMACSHA512

C. DPAPI blob Algorithms are written in the blob itself.

Classic DPAPI Flow: Getting the system's secrets (easy)

Cached Logons: It used to be like this...

Windows 2003 / XP

The encryption algorithm is RC4.
The hash is used to verify authentication is calculated as follows:

```
DCC1 = MD4(MD4(Unicode(password)) .
LowerUnicode(username))
is
DCC1 = MD4(hashNTLM . LowerUnicode(username))
```

Usage in the attack

Before the attacks facilitated by pass-the-hash, we can only rejoice the "salting" by the username.

There are a number pre-computed tables for users as Administrator facilitating attacks on these hashes.

Cached Logons

Windows Vista / 2008 +

The encryption algorithm is AES128.

The hash is used to verify authentication is calculated as follows:

MSDCC2 = PBKDF2(HMAC-SHA1, Iterations, DCC1, LowerUnicode(username))

with DCC 1 calculated in the same way as for 2003 / XP.

Usage in the attack

There is actually not much of a difference with XP / 2003! No additional salting.

PBKDF2 introduced a new variable: the number of iterations SHA1 with the same salt as before (username).

Sysmon stores a hash base

Getting the: Cached Data

MSDCC2

1. bootkey: classes from HKLM\SYSTEM\CCS\Control\Lsa + [class names for: Data, GBG, JD, Skew1] (+arrays' permutations) int[] permutationBootKey = new int[] { 0x8, 0x5, 0x4, 0x2, 0xb, 0x9, 0xd, 0x3, 0x0, 0x6, 0x1, 0xc, 0xe, 0xa, 0xf, 0x7 };
2. PolEKList: HKLM\SECURITY\Policy\PolEKList [default value]
3. lsakey: AES_DECRYPT(key, data) -> AES(bootkey, PolEKList)
4. NL$KM secret: HKLM\SECURITY\Policy\Secrets\NL$KM
5. nlkm_decrypted: AES_DECRYPT(lsakey, NL$KM secret)
6. Cache_Entry{id} -> HKLM\SECURITY\Cache\NL${id}
7. cache_entry_decrypted -> AES_DECRYPT(nlkm_decrypted, Cache_Entry{id})

Encrypted Cached Credentials: Legend

```
        0  1  2  3  4  5  6  7  8  9  A  B  C  D  E  F   0123456789ABCDEF
0000h: 10 00 0A 00 10 00 1C 00 00 00 00 00 00 00 00 00   ................
0010h: 8B 04 00 00 01 02 00 00 02 00 00 00 0A 00 18 00   ‹...............
0020h: 26 C7 A8 43 88 7F D0 01 04 00 01 00 01 00 00 00   &Ç¨Cˆ.Ð........
0030h: 01 00 0A 00 10 00 00 00 10 00 00 00 12 00 24 00   ..............$.
0040h: 4A 4F 26 05 63 9B C3 22 9F 97 77 E6 B0 CD 52 BA   JO&.c›Ã"Ÿ—wæ°ÍR°
0050h: C0 76 14 67 D6 68 37 04 87 72 95 DC 19 6D 26 90   Àv.gÖh7.‡r•Ü.m&.
0060h: 15 5C 25 C7 A8 17 05 7B A3 D0 5C 6F 3C A7 82 4A   .\%Ç¨..{£Ð\o<§‚J
0070h: 52 72 D1 B6 1F 91 6B B7 9C D2 20 9A 1B 25 ED A0   RrÑ¶.‘k·œÒ š.%í
0080h: 68 E5 4D 3E 42 F6 C4 BA 68 A1 BD CB 5A 73 4A 89   håM>BöÄºh¡½ËZsJ‰
0090h: 07 C7 E2 C5 50 20 4E D6 CD 02 BA BB E6 E9 CA F0   .ÇâÅP NÖÍ.º»æéÊð
00A0h: 8C 17 4E CF 60 F7 90 D3 37 FB 30 4B C3 95 B7 02   Œ.NÏ`÷.Ó7û0KÃ•·.
00B0h: D6 38 75 63 D2 0F 15 AD 3A C4 32 53 D5 8B 66 7D   Ö8ucÒ..-:Ä2SÕ‹f}
00C0h: 9D FB 5D AA 30 7E B7 A5 F5 9B 57 32 D9 47 EE EE   .û]ª0~·¥õ›W2ÙGîî
00D0h: 5C 07 6C 3B 64 78 A7 B1 78 C2 EA F5 98 A8 CB B1   \.l;dx§±xÂêõ˜¨Ë±
00E0h: DD 34 92 00 93 9F 65 9D 38 E7 7B F9 69 53 97 50   Ý4’.“Ÿe.8ç{ùiS—P
00F0h: CB 82 49 38 CF B4 CA F9 4B EB D8 8E 4C D4 6D CE   Ë‚I8Ï´ÊùKëØŽLÔmÎ
0100h: 09 7E 6F F6 65 49 C6 9F 61 8D 4A 16 24 3A 40 CB   .~oöeIÆŸa.J.$:@Ë
0110h: CC 3C D8 FD FC 91 6B E5 84 5E 68 9C 69 D7 B4 FD   Ì<Øýü‘kå„^hœi×´ý
0120h: 62 44 8D 23 E8 0A 1E BE BB 34 EB 81 23 FE E3 0E   bD.#è..¾»4ë.#þã.
0130h: 76 55 9E 63 9E DE 57 DC 0C 60 BE A8 53 AF BD AA   vUžcžÞWÜ.`¾¨S¯½ª
0140h: AB 3F ED 7A EE B4 62 50 EC E1 B8 B1 8F 9E A6 2B   «?ízî´bPìá¸±.ž¦+
0150h: 9B 85 71 63 D9 6C 66 09 C2 70 DC 63 E6 22 E8 08   ›…qcÙlf.ÂpÜcæ"è.
0160h: A4 55 5F 36 C2 64 1E 2B B8 80 6A A5 AC 17 92 41   ¤U_6Âd.+¸€j¥¬.’A
0170h: 3C 21 2E DF CC EA 75 9E 99 31 C4 D6 8C AF C7 04   <!.ßÌêuž™1ÄÖŒ¯Ç.
0180h:
```

Name	Value	Start	Size	Color	Comment
struct Header h		0h	96	Fg: Bg:	
ushort uname_len	16	0h	2	Fg: Bg:	
ushort domain_len	10	2h	2	Fg: Bg:	
ushort mail_nick_len	16	4h	2	Fg: Bg:	
ushort cn_len	28	6h	2	Fg: Bg:	
ushort u1	0	8h	2	Fg: Bg:	
ushort logon_script_len	0	Ah	2	Fg: Bg:	
ushort profile_path_len	0	Ch	2	Fg: Bg:	
ushort home_dir_len	0	Eh	2	Fg: Bg:	
uint user_sid	1163	10h	4	Fg: Bg:	
uint primary_group_id	513	14h	4	Fg: Bg:	
uint u2	2	18h	4	Fg: Bg:	
ushort group_sids_len	10	1Ch	2	Fg: Bg:	
ushort domain_netbios_name...	24	1Eh	2	Fg: Bg:	
FILETIME last_local_logon	04/25/2015 18:47:22	20h	8	Fg: Bg:	
ushort u3	4	28h	2	Fg: Bg:	
ushort u4	1	2Ah	2	Fg: Bg:	
uint u5	1	2Ch	4	Fg: Bg:	
ushort u6	1	30h	2	Fg: Bg:	
ushort u7	10	32h	2	Fg: Bg:	
uint u8	16	34h	4	Fg: Bg:	
uint u9	16	38h	4	Fg: Bg:	
ushort domain_name_len	18	3Ch	2	Fg: Bg:	
ushort email_len	36	3Eh	2	Fg: Bg:	
byte iv[16]	JO&|c›Ã"Ÿ—wæ°ÍR°	40h	16	Fg: Bg:	

Encrypted Cached Credentials

DK = PBKDF2(PRF, Password, Salt, c, dkLen)

Microsoft’s implementation: MSDCC2= PBKDF2(HMAC-SHA1, DCC1, username, 10240, 16)

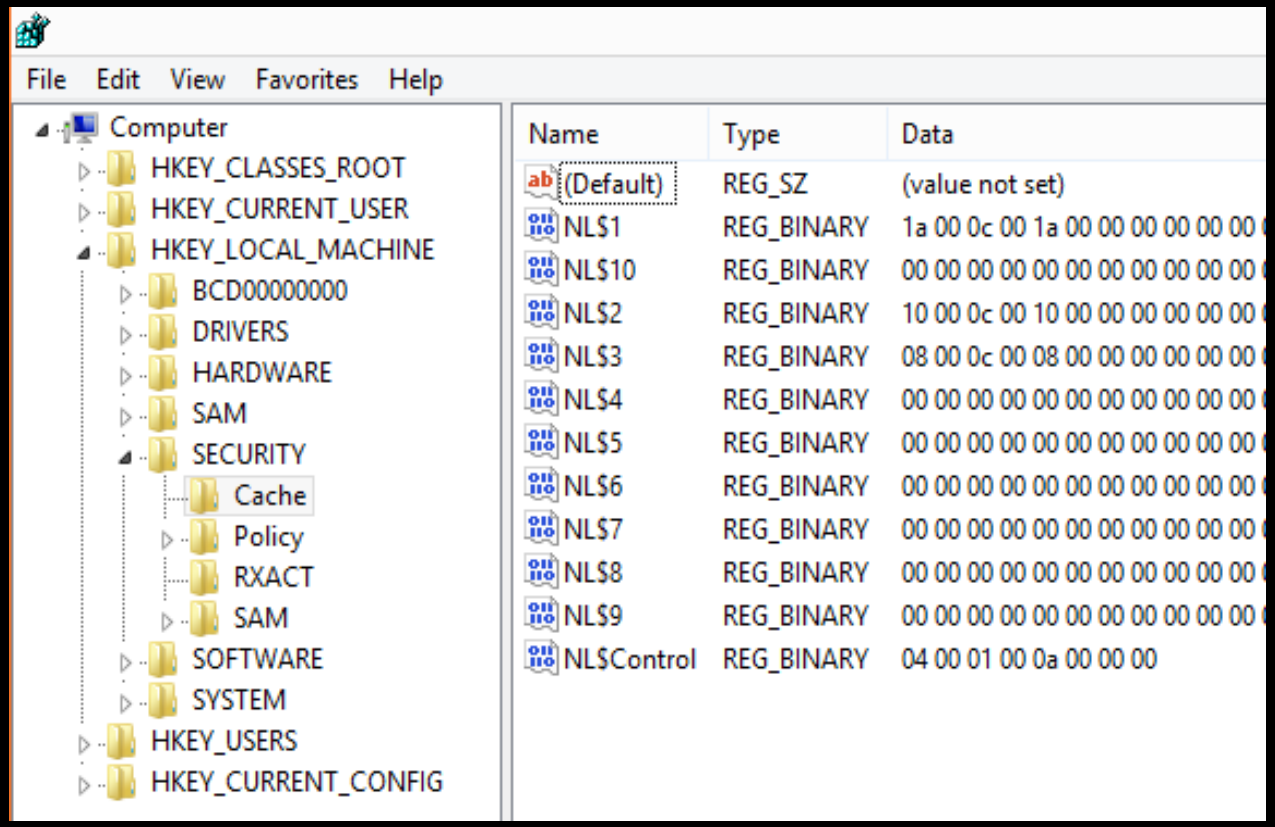
Cached Logons: Iterations

The number of iterations in PBKDF2 is configurable through the registry:

HKEY_LOCAL_MACHINE\SECURITY\Cache DWORD (32) NL$IterationCount

If the number is less than 10240, it is a multiplier by 1024 (20 therefore gives 20480 iterations)

If the number is greater than 10240, it is the number of iterations (rounded to 1024)



DPAPI + AD

CLIENT

Local LSASS process (local masterkey can't be decrypted)

DPAPI

CryptUnprotectData()

```
01 00 00 00  d0 8c 9d df  01
15 d1 11  8c 7a 00 c0
4f c2 97 eb  01 00 00 00
......
```

DPAPI-PROTECTED BLOB

RPC Call

BackupKey(masterkey)

Decrypted masterkey

Scenario: offline changed user password or local masterkey can't be decrypted

AD SERVER

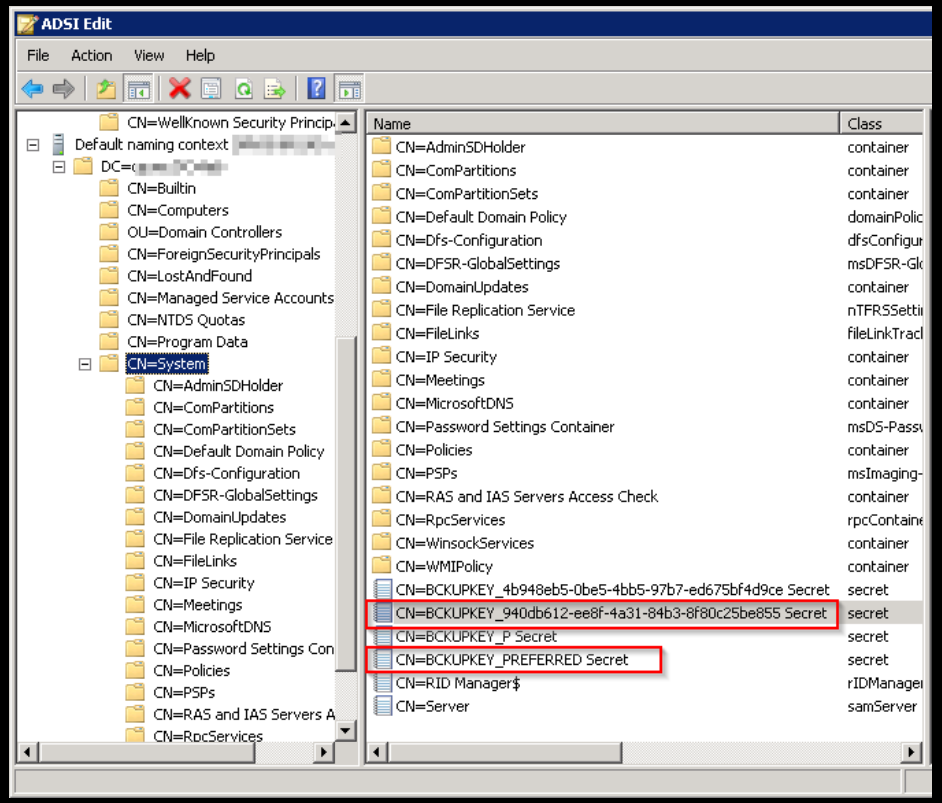
LSASS process

G$BCKUPKEY_PREFERRED

G$BCKUPKEY_940db612-ee8f-4a31-84b3-8f80c25be855

Classic DPAPI Flow: getting the user's secrets

Retrieving Golden Key from LSA – CQURE's way



AD secret? HOW?!

? ? ? ? ? ? ? ?

LSASS.EXE MEMORY

LSASRV.DLL

G$BCKUPKEY_PREFERRED

G$BCKUPKEY_940db612-ee8f-4a31-84b3-8f80c25be855

LsaRetrievePrivateData()

RSA private/public key pair

CQLsassSecretsDumper

GoldenKey.pfx

CQURE

DPAPI-AD: How did we do it?

DomainKey contains some
GUID and
256-byte len secret – RSA??

Dude, look in the AD…

```
02 00 00 00 00 00 00 00 00 00 00 00 33 00 36 00   ............3.6.
64 00 63 00 65 00 30 00 33 00 66 00 2D 00 36 00   d.c.e.0.3.f.-.6.
63 00 35 00 65 00 2D 00 34 00 65 00 39 00 38 00   c.5.e.-.4.e.9.8.
2D 00 38 00 33 00 63 00 38 00 2D 00 32 00 35 00   -.8.3.c.8.-.2.5.
33 00 33 00 61 00 30 00 34 00 31 00 39 00 62 00   3.3.a.0.4.1.9.b.
37 00 64 00 00 00 00 00 00 00 00 00 00 00 00 00   7.d.............
88 00 00 00 00 00 00 00 68 00 00 00 00 00 00 00   ˆ.......h.......
00 00 00 00 00 00 00 00 74 01 00 00 00 00 00 00   ........t.......
02 00 00 00 35 77 9B 32 0C 12 0C EF 04 AB D4 84   ....5w›2...ï.«Ô„
E7 A0 80 A4 C0 5D 00 00 09 80 00 00 03 66 00 00   ç €¤À]...€...f..
C7 29 95 2B E0 E3 3D 29 3C 56 EC 3B BB 11 09 F1   Ç)•+àã=)<Vì;»..ñ
BA A4 D0 E5 8C 49 B6 B7 C2 5A 11 D8 86 3C C4 65   º¤ÐåŒI¶·ÂZ.Ø†<Äe
2C 0D 7D 1D 1C B6 B2 91 69 1D 77 F1 7A E5 29 38   ,.}..¶²‘i.wñzå)8
90 D6 FA 4F 13 96 10 97 68 FE 08 98 26 35 26 F0   .ÖúO.–.—hþ.˜&5&ð
A6 E7 BB 03 5D 65 BF B6 8E 66 0D 95 E8 C2 E7 52   ¦ç».]e¿¶Žf.•èÂçR
E5 29 97 65 E6 E9 FE 09 32 90 70 8E F3 07 F7 1F   å)—eæéþ.2.pŽó.÷.
EE 84 AB 88 B8 D3 A2 04 02 00 00 00 76 41 19 68   î„«ˆ¸Ó¢.....vA.h
B9 96 9D A2 E0 C2 DC D8 8F A0 2D F5 C0 5D 00 00   ¹–.¢àÂÜØ. –õÀ]..
09 80 00 00 03 66 00 00 16 6B FA F7 7A EA A6 CD   .€...f...kú÷zê¦Í
B1 BB C0 A7 6F 58 02 03 25 FD D8 DD A6 3C D1 ED   ±»À§oX..%ýØÝ¦<Ñí
2E CE E0 17 5C B5 03 F4 E3 A1 F7 D2 37 85 65 DE   .Îà.\µ.ôã¡÷Ò7…eÞ
E7 70 76 64 4B C4 76 17 50 0B 4C AD 37 4B 8C 74   çpvdKÄv.P.L-7KŒt
22 CD BE 91 C0 7D A3 A7 F3 2A 59 9D 52 0C F3 97   "Í¾‘À}£§ó*Y.R.ó—
02 00 00 00 00 01 00 00 58 00 00 00 12 B6 0D 94   ........X....¶.”
8F EE 31 4A 84 B3 8F 80 C2 5B E8 55 3C E3 C6 C2   .î1J„³.€Â[èU<ãÆÂ
BD 1F 88 A3 CE 4D EF FC CC 23 56 78 E5 58 E4 A9   ½.ˆ£ÎMïüÌ#VxåXä©
D9 78 4A FA 47 B2 21 89 F0 F4 68 5E 7B 93 FB 27   ÙxJúG²!‰ðôh^{“û'
6E 19 EF B5 52 6E 03 50 A9 8F 1A 5B 99 1E F5 01   n.ïµRn.P©..[™.õ.
39 59 9F 1D 00 A0 C2 6E 9E 48 25 B3 20 0E DC C1   9YŸ.. ÂnžH%³ .ÜÁ
4D 9C 3E E3 68 6E 47 8D 32 57 03 5B CB BE A0 7F   Mœ>ãhnG.2W.[Ë¾ .
E5 4F 27 C7 EB 82 0C E2 5E 00 9A CA 6F 0D ED 10   åO'Çë‚.â^.šÊo.í.
FD EB 12 01 FB 9B 60 1D E2 38 4F FD 58 46 69 AC   ýë..û›`.â8OýXFi¬
D3 75 2E 79 80 D8 72 99 C0 25 20 E0 93 DD DD 2E   Óu.y€Ør™À% à“ÝÝ.
5B 1A EA 8C 4C F1 51 99 13 70 F8 F0 8A 4F F3 57   [.êŒLñQ™.pøðŠOóW
AD 07 DD 5B C5 C3 24 3B 01 9C 8D B7 67 65 85 B3   -.Ý[ÅÃ$;.œ.·ge…³
70 B3 54 80 C1 68 19 31 C8 3F 0A 3F 69 FC 7A E0   p³T€Áh.1È?.?iüzà
23 F2 75 68 05 BF 3C 8A A0 CC 73 B4 C4 69 A7 C5   #òuh.¿<Š Ìs´Äi§Å
FB 53 B5 91 73 12 A7 41 E6 45 06 AC F3 69 54 B7   ûSµ‘s.§AæE.¬óiT·
A1 F7 C3 8C E6 F8 23 88 2C C9 E2 C0 5D 08 EA 47   ¡÷ÃŒæø#ˆ,ÉâÀ].êG
BA 80 16 06 D8 1D 28 37 21 DE B2 79 8B 34 D8 8D   º€..Ø.(7!Þ²y‹4Ø.
51 F0 FD A6 D2 E1 DC 9A 2C 9B 38 4A B4 2F DA 02   Qðý¦ÒáÜš,›8J´/Ú.
67 CC 0F 8A EC 83 A9 9A AA B0 C9 39 95 86 B3 27   gÌ.Šìƒ©šªºÉ9•†³'
A0 E7 43 14 12 4F 2D A7 A9 36 49 16 3E E9 BB AD    çC..O-§©6I.>é»-
47 06 C6 18 56 3F 9E 02 1F 0F 93 33 0D 82 D9 98   G.Æ.V?ž...“3.‚Ù˜
DE BA BA 2F 7E ED FC 70 79 DA 4C 7A C7 60 C7 36   Þºº/~íüpyÚLzÇ`Ç6
54 23 C5 51 7A D1 94 C8 2C E9 85 05 5B 2C 6F 4F   T#ÅQzÑ”È,é….[,oO
81 B1 06 41                                       .±.A
```

Name	Value	Start				Comment
struct MasterKeyFile mkf		0h				
uint version	2	0h				
uint unknown1	0	4h				
uint unknown2	0	8h	4h			
wchar_t guid[36]	36dce03f-6c5e-4e98-83c8-2533a0419b7d	Ch	48h			
uint unknown3	0	54h	4h			
uint unknown4	0	58h	4h			
uint policy	0	5Ch	4h			
quad masterkeyLen	136	60h	8h			
quad backupkeyLen	104	68h	8h	Fg:	Bg:	
quad credhistLen	0	70h	8h	Fg:	Bg:	
quad domainkeyLen	372	78h	8h	Fg:	Bg:	
struct MasterKey masterkey		80h	88h	Fg:	Bg:	
uint version	2	80h	4h	Fg:	Bg:	
byte iv[16]	5w›2□□□ï□«Ô„ç €¤	84h	10h	Fg:	Bg:	
uint rounds	24000	94h	4h	Fg:	Bg:	
uint hashAlgo	32777	98h	4h	Fg:	Bg:	
uint cipherAlgo	26115	9Ch	4h	Fg:	Bg:	
byte cipherText[104]	Ç)•+àã=)<Vì;»□ ñº¤ÐåŒI¶·ÂZ□Ø†<Ä…	A0h	68h	Fg:	Bg:	
struct MasterKey backupkey		108h	68h	Fg:	Bg:	
struct DomainKey domainkey		170h	174h	Fg:	Bg:	
uint version	2	170h	4h	Fg:	Bg:	
uint secretLen	256	174h	4h	Fg:	Bg:	
uint accessCheckLen	88	178h	4h	Fg:	Bg:	
struct GUID guidKey	940db612-ee8f-4a31-84b3-8f80c25be855	17Ch	10h	Fg:	Bg:	
byte encryptedSecret[256]	ŒãÆÂ½□ˆ£ÎMïüÌ#VxåXä©ÙxJúG²!‰ðô…	18Ch	100h	Fg:	Bg:	
byte accessCheck[88]	´/Ú□gÌ□Šìƒ©šªºÉ9•†³' çC□□O-§©6I□…	28Ch	58h	Fg:	Bg:	

Demo:
What about KeePass?

DPAPI in pictures Example: KeePass ProtectedUserKey.bin

```
0000h: 01 00 00 00 D0 8C 9D DF 01 15 D1 11 8C 7A 00 C0  ....ÐŒ.ß..Ñ.Œz.À
0010h: 4F C2 97 EB 01 00 00 00 9E 4F 95 AE CF 21 62 46  OÂ—ë....žO•®Ï!bF
0020h: AC EA 6B E2 FC FC 23 B3 00 00 00 00 02 00 00 00  ¬êkâüü#³........
0030h: 00 00 10 66 00 00 00 01 00 00 20 00 00 00 5E 67  ...f.... ...^g
0040h: 54 64 F4 D5 D7 E4 CB 14 23 53 B4 8E 4B 44 61 F9  TdôÕ×äË.#S´ŽKDaù
0050h: CE E3 76 9D F4 25 08 23 44 DC 35 32 C2 70 00 00  Îãv.ô%.#DÜ52Âp..
0060h: 00 00 0E 80 00 00 00 02 00 00 20 00 00 00 D6 BD  ...€.... ...Ö½
0070h: 40 A5 3D 14 B7 6A 84 54 56 6E 6C 03 B8 9D 8D DA  @¥=.·j„TVnl.¸..Ú
0080h: D0 AF C8 1B F2 16 26 E4 1C F3 A3 FA 10 1B 50 00  Ð¯È.ò.&ä.ó£ú..P.
0090h: 00 00 2F C6 5A 86 0F 66 04 BA 25 D5 C2 A3 89 EB  ../ÆZ†.f.º%ÕÂ£‰ë
00A0h: 2C 33 E1 38 6E D6 41 0E D3 E9 E7 E3 B7 5D B2 E8  ,3á8nÖA.Óéçã·]²è
00B0h: B4 3F 79 36 0F 6E 1F D1 67 D0 B7 06 D8 C1 20 25  ´?y6.n.ÑgÐ·.ØÁ %
00C0h: C1 B5 DF 11 9F DD FF A4 CF BC A6 3E 20 A5 C9 4C  Áµß.ŸÝÿ¤Ï¼¦> ¥ÉL
00D0h: AA D4 C3 16 4F 68 C7 AB B0 66 80 E5 DA 2D 6E A0  ªÔÃ.OhÇ«°f€åÚ-n 
00E0h: CA 35 40 00 00 00 1D 0D 07 C3 22 BD 40 6E EB 58  Ê5@....Ã"½@nëX
00F0h: 54 C7 B8 9D 7E 1E 6A 93 41 59 EB B3 8E 4A 66 72  TÇ¸.~.j"AYë³ŽJfr
0100h: 5F 43 0A D9 40 CC 37 09 19 AF 6F 7C 91 21 1F 60  _C.Ù@Ì7..¯o|'!.`
0110h: 59 35 2E 20 01 CE 38 F7 E4 5C 0D 8A 8B 28 80 11  Y5. .Î8÷ä\.Š‹(€.
0120h: 84 84 AB 24 91 52                                „„«$'R
```

Name	Value	Start	Size	Color	Comment
struct DPAPIBlob blob		0h	126h	Fg: Bg:	
uint version	1	0h	4h	Fg: Bg:	
struct GUID provider	df9d8cd0-1501-11d1-8c7a-00c04fc297eb	4h	10h	Fg: Bg:	
uint mkversion	1	14h	4h	Fg: Bg:	
struct GUID mkguid	ae954f9e-21cf-4662-acea-6be2fcfc23b3	18h	10h	Fg: Bg:	
uint flags	0	28h	4h	Fg: Bg:	
uint descriptionLen	2	2Ch	4h	Fg: Bg:	
wstring description[1]		30h	2h	Fg: Bg:	
uint cipherAlgo	26128	32h	4h	Fg: Bg:	
uint keyLen	256	36h	4h	Fg: Bg:	
uint saltLen	32	3Ah	4h	Fg: Bg:	
byte salt[32]	^gTdôÕ×äË□#S´ŽKDaùÎãv□ô%□#DÜ5...	3Eh	20h	Fg: Bg:	
uint strongLen	0	5Eh	4h	Fg: Bg:	
uint hashAlgo	32782	62h	4h	Fg: Bg:	
uint hashLen	512	66h	4h	Fg: Bg:	
uint hmacLen	32	6Ah	4h	Fg: Bg:	
byte hmac[32]	Ö½@¥=□·j„TVnl□¸□□ÚÐ¯È□ò□&ä□ó...	6Eh	20h	Fg: Bg:	
uint cipherTextLen	80	8Eh	4h	Fg: Bg:	
byte cipherText[80]	/ÆZ†□f□º%ÕÂ£‰ë,3á8nÖA□Óéçã·]²è...	92h	50h	Fg: Bg:	
uint signLen	64	E2h	4h	Fg: Bg:	
byte sign[64]	□ □Ã"½@nëXTÇ¸□~□j"AYë³ŽJfr_C Ù...	E6h	40h	Fg: Bg:	

The master password for KeePass files encrypted & stored as cipherText (80 bytes)

DPAPI blob: Legend

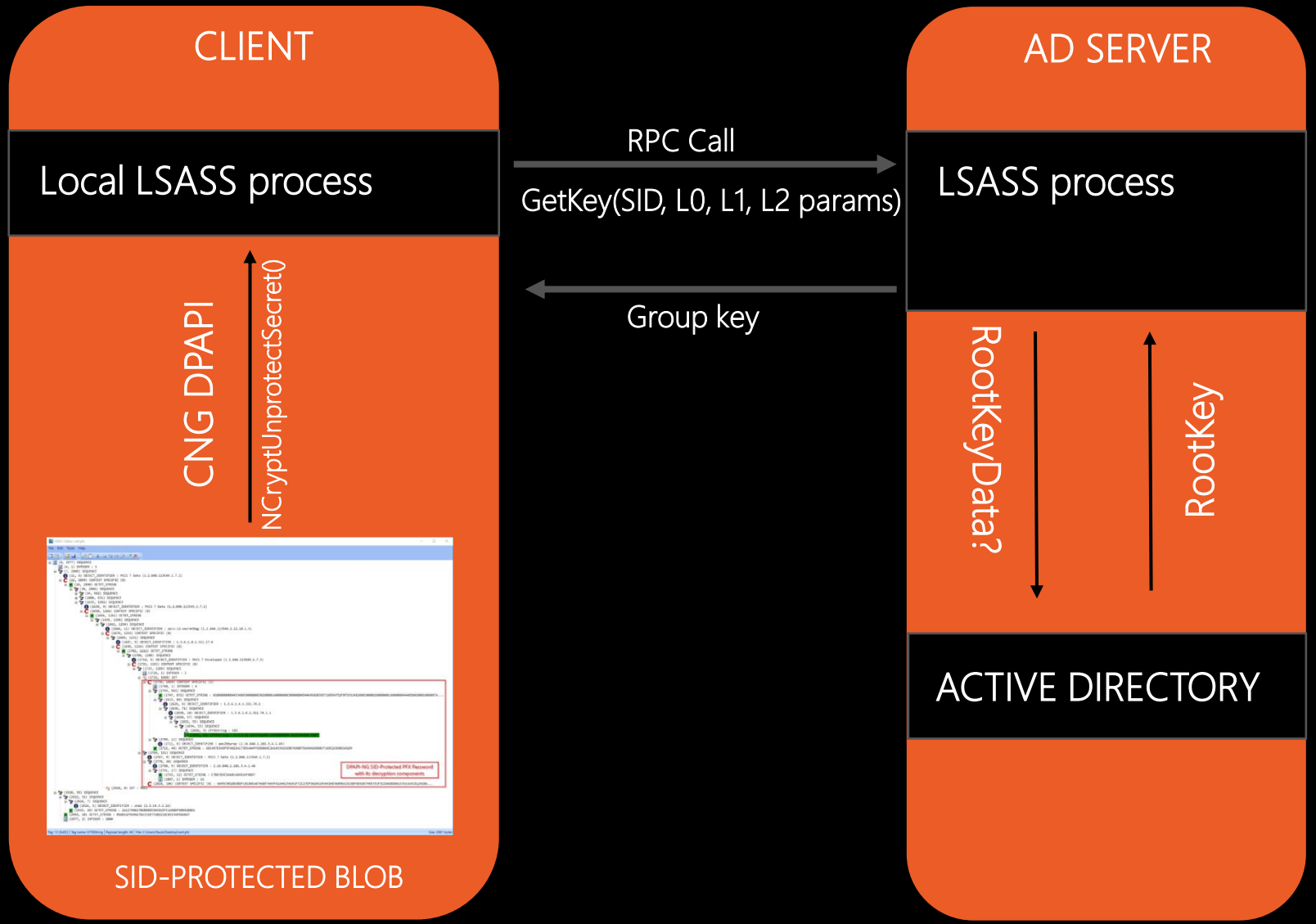
Demo:
What about RDP Connections?

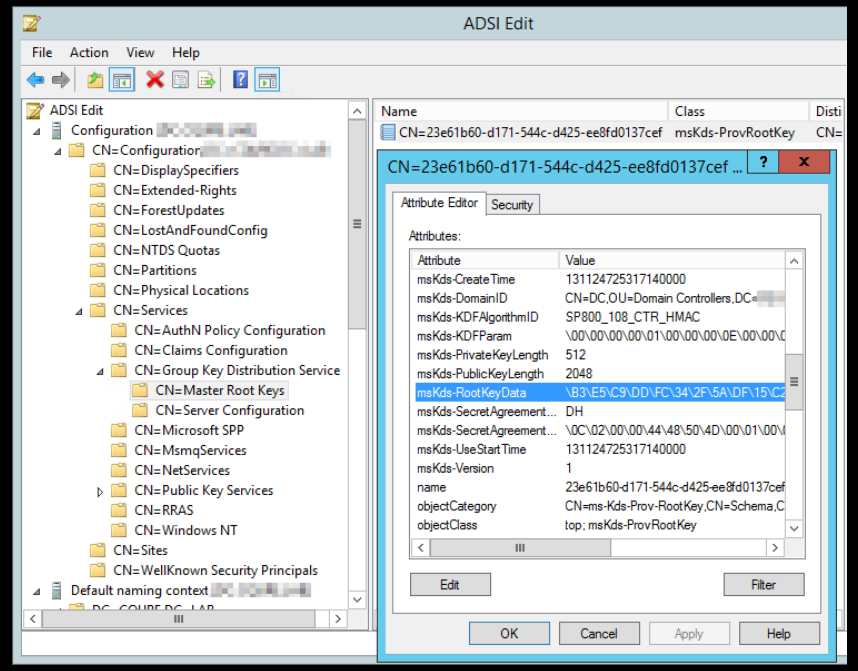
Getting the: DPAPI-NG Secrets

DPAPI-NG

A. RootKey Algorithms Key derivation function:
SP800_108_CTR_HMAC (SHA512) Secret agreement: Diffie-Hellman
B. DPAPI blob Key derivation: KDF_SP80056A_CONCAT
After getting the key, there is a need for decryption: Key wrap algorithm: RFC3394 (KEK -> CEK) Decryption: AES-256-GCM (CEK, Blob)

DPAPI-NG: Data Encryption Flow





DPAPI-NG: Protected Data Encoded as ASN.1 blob

```
(0, 478) SEQUENCE
  (4, 9) OBJECT_IDENTIFIER : PKCS 7 Enveloped (1.2.840.113549.1.7.3)
  (15, 463) CONTEXT SPECIFIC (0)
    (19, 459) SEQUENCE
      (23, 1) INTEGER : 2
      (26, 378) SET
        (30, 374) CONTEXT SPECIFIC (2)
          (34, 1) INTEGER : 4
          (37, 312) SEQUENCE
            (41, 262) OCTET_STRING : 01000000D08C9DDF0115D1118C7A00C04FC297EB01000000C38A3C14E312944EB51BFBC9FC3E37510000
            (307, 44) SEQUENCE
              (309, 9) OBJECT_IDENTIFIER : 1.3.6.1.4.1.311.74.1
              (320, 31) SEQUENCE
                (322, 10) OBJECT_IDENTIFIER : 1.3.6.1.4.1.311.74.1.8
                (334, 17) SEQUENCE
                  (336, 15) SEQUENCE
                    (338, 13) SEQUENCE
                      (340, 5) UTF8String : LOCAL
                      (347, 4) UTF8String : user
          (353, 11) SEQUENCE
            (355, 9) OBJECT_IDENTIFIER : aes256wrap (2.16.840.1.101.3.4.1.45)
            (366, 40) OCTET_STRING : 2CD21C53E6DA47B6371DAF77471DB800F13EF8F41F8F83A049EEF4C552A9C5B8DA72D9BC4BE0B8A3
    (408, 72) SEQUENCE
      (410, 9) OBJECT_IDENTIFIER : PKCS 7 Data (1.2.840.113549.1.7.1)
      (421, 30) SEQUENCE
        (423, 9) OBJECT_IDENTIFIER : 2.16.840.1.101.3.4.1.46
        (434, 17) SEQUENCE
          (436, 12) OCTET_STRING : E8F7BCA9937A67C74F3B5E0B
          (450, 1) INTEGER : 16
      (453, 27) CONTEXT SPECIFIC (0) : 13F67D65A52DBD6EC2508BBD5B87A4E71BDF4464A49F55A2106E5F
```

Protection descriptor: LOCAL=user

- KEK (Key Encryption Key) stored as DPAPI blob
- Forced by protection descriptor LOCAL=user
- Key Wrap (RFC3394) contains encrypted CEK (Content Encryption Key)
- Data encrypted by CEK

Summary

Summary

#1
Implement MFA across your organization

#2
Consider adopting a Zero-Trust policy

#3
Learn where your credentials live and how can your secrets get extracted

#4
Use solid and up-to-date security measures

#5
Invest in awareness and specialist training for employees

#6
Try to stay up-to-date with current news and trends from the world of cybersecurity

Q&A time!

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Username: student
Password: CQUREAcademy#123!

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DPAPI and DPAPI-NG:
Decrypting All Users' Secrets and PFX Passwords

Paula Januszkiewicz

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Decrypting All Users' Secrets and PFX Passwords

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Paula Januszkiewicz

CEO and Founder

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